Import Exposure and the April 2025 Reciprocal-Tariff Shock: A Paired Event Study of the "Liberation Day" Crash and the April 9 Pause

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Abstract

On April 2, 2025 the United States announced sweeping "reciprocal" tariffs; the equity market fell about 10.7% over the next two trading days—its worst two-day loss since March 2020—before the largest one-day gain since 2008 (+9.7%) when a 90-day pause was announced on April 9. We use this paired, opposite-signed pair of events to test whether firms exposed to *imported inputs* were repriced by the tariff news. For 4,275 U.S. common stocks we compute market-model cumulative abnormal returns (CARs)

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`). A note on author order and status: the student is listed first here as a program convention, but in finance and economics published author order is conventionally alphabetical; all authorship on this draft is tentative and specific to the ASSIP 2026 program at this stage.

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over the shock (April 3–4) and the pause (April 9), and relate them to the imported-input intensity of each firm's industry, built from the BEA Summary Import Matrix (imported intermediate inputs ÷ industry gross output). Conditional on size and market beta, a one-standard-deviation increase in imported-input intensity is associated with a 0.73 percentage-point *lower* abnormal return on the shock ($t = -2.35$) and a 0.60pp *higher* abnormal return on the pause ($t = 1.78$); among non-microcap firms the pattern is opposite-signed and nearly symmetric (-0.77pp and $+0.77$pp, both $t \approx 2.1$), and the shock-plus-pause round-trip is statistically zero. The effect is a *conditional* one—it is absent in the raw cross-section, which is dominated by size and beta—and it is partly anticipated: exposure loads negatively in the pre-announcement window ($t = -2.22$). We then stress-test the result with design-based methods. A falsification on generic big-market-move days shows the exposure loading is directionally coherent across the paired tariff events rather than a residual-beta artifact, but it is not silent on every non-tariff day. Most importantly, because exposure varies at the *industry* level (73 NAICS-3 groups), we conduct randomization inference that permutes industry exposure across industries: the opposite-signed spread and the shock loading are significant only at the 10% level ($p \approx 0.08$–0.10) and the pause alone is insignificant—so conventional cluster-robust t-statistics overstate the precision available from an industry proxy. A minimum-detectable-effect analysis confirms the design sits at the edge of its resolving power for effects of this size. The honest reading is *suggestive, directionally consistent evidence of an industry-level tariff-exposure repricing whose statistical strength is limited by the coarseness of the industry measure*—which is precisely the case for the hand-collected, firm-level 10-K sourcing data the companion student task is built to supply.

JEL: F13, F14, G14, G12. **Keywords:** tariffs, trade war, imported inputs, event study, abnormal returns, randomization inference.

1 Introduction

Between April 2 and April 9, 2025, U.S. equities completed a violent round trip. After the close on April 2 ("Liberation Day"), the administration announced a schedule of country-specific "reciprocal" tariffs far larger than markets had expected; over April 3 and April 4 the aggregate market fell about 10.7%, the worst two-day loss since the COVID-19 crash. Selling continued through April 8. Then, roughly midday on April 9, a 90-day pause on most of the reciprocal tariffs (excluding China) was announced, and the market rose 9.7% in a single session—the largest one-day gain since October 2008. The speed and symmetry of this episode create an unusually clean laboratory: two large, opposite-signed shocks to the *same* policy, six trading days apart, with little else changing.

We use this laboratory to ask a specific cross-sectional question. Tariffs raise the cost of imported intermediate inputs. Firms that rely heavily on imported inputs therefore face a cost shock when tariffs are imposed and relief when they are paused. Did the equity market price this? Concretely: *did firms in high imported-input-intensity industries underperform on the April 2–4 shock and rebound on the April 9 pause?* Because the two events push in opposite directions, a genuine tariff-exposure channel should generate an *opposite-signed* loading on the same exposure measure—a pattern that is much harder to explain with a generic risk factor than a single-event correlation would be.

Our exposure measure is built from real input-output data. Using the Bureau of Economic Analysis (BEA) 2023 Summary Import Matrix, we compute, for each of 66 BEA industries, the value of imported intermediate inputs used by that industry divided by its gross output. The measure is highest for petroleum and coal products (24.7%), motor vehicles (18.3%), primary metals (15.6%), machinery (12.1%), and chemicals (10.8%), and lowest for services such as legal services, real estate, and software—exactly the ranking one would expect for input-cost exposure to tariffs. We map this industry measure to firms through their NAICS code and merge it to market-model CARs for the CRSP universe.

Our findings are three. First, in the *raw* cross-section imported-input intensity has no clean relationship with the shock CAR: the univariate coefficient is small and wrongly signed ($t = 1.20$), because import-intensive industries are disproportionately large, high-beta firms and those two characteristics dominate the unconditional cross-section. Second, and centrally, *once we condition on size and market beta the hypothesized paired pattern appears.* A one-standard-deviation increase in imported-input intensity is associated with a 0.73pp lower shock CAR ($t = -2.35$) and a 0.60pp higher pause CAR ($t = 1.78$); over the full April 3–8 sell-off the shock effect is -1.34pp ($t = -2.48$). Excluding microcaps sharpens the symmetry to -0.77pp on the shock and $+0.77$pp on the pause, both significant at 5% under conventional inference, and the shock-plus-pause round-trip is a precise zero ($t = -0.46$): the pause reversed the shock's exposure effect. Third, we are candid that the episode is not a laboratory-clean event. Imported-input intensity also loads negatively in the April week *before* the announcement ($t = -2.22$), because tariff news was already flowing—auto tariffs were announced March 26 and the reciprocal schedule was pre-trailed—so part of the repricing preceded April 2.

The core of this revision is to subject that headline to design-based scrutiny rather than stopping at the conventional t-statistics, because the exposure measure varies at the industry level and industry proxies are known to inflate apparent precision (Cameron et al., 2008; Young, 2019). We add four steps. (1) *Heterogeneity* (Section 6, Table 7): we interact the exposure loading with observable pre-event moderators. The clean economic prediction—that the effect concentrates in input-heavy (high-COGS/sales) firms—is *not* confirmed at the industry level, which is itself informative: the industry proxy cannot see which firms within an industry actually source abroad. (2) *Falsification* (Section 7, Table 8): we recompute exposure loadings on the largest *non-tariff* market-move days of 2024 using the same market model. The loadings there are smaller and sign-inconsistent, so the tariff pattern is distinguished by its coherent opposite-signed pairing on the same measure rather than by any single-day loading—but exposure is not literally silent on every big day, an honest qualifier. (3) *Randomization inference* (Section 8, Table 9, Figure 3): we permute the 73 industry-level intensity values across industries 1,000 times and rebuild the loadings. The shock loading and the opposite-signed spread $D = \hat{\beta}_{\text{pause}} - \hat{\beta}_{\text{shock}}$ are significant only at the 10% level ($p \approx 0.08$–0.10); the pause alone is insignificant ($p = 0.16$). The clustered-SE t-statistics therefore overstate the evidence available from an industry proxy. (4) *Power* (Section 9, Table 10): the design's minimum detectable effect at 80% power is about 0.9pp per SD, just above the estimated loadings—so the marginal significance under randomization inference is a resolution problem, not a sign problem.

Taken together, the evidence is *suggestive and directionally consistent* but not decisive: the market appears to have priced imported-input exposure across the shock and the pause in a sign-coherent way, yet the statistical strength attainable from a 73-industry proxy is limited. This is not a weakness to be hidden but the paper's central motivation. The identifying variation that would sharpen both the point estimate and the inference lives *within* industries—which apparel firm sources from Vietnam versus reshored to Mexico; which retailer imports finished goods that the input-output measure misses entirely—and it can only be recovered by reading filings. That firm-level, hand-collected 10-K sourcing measure, and the country-level April-2-versus-April-9 tariff map that lets a China-sourcing firm and a Vietnam-sourcing firm react differently to the pause, are the companion student contribution; we deliberately restrict every test in this paper to *observable* industry- and firm-level characteristics so that the firm-level sourcing dimension is reserved for that sharper test.

We contribute to two literatures. Relative to the trade-war economics literature (Amiti et al., 2019; Fajgelbaum et al., 2020; Handley and Limão, 2017), which documents that the 2018–2019 tariffs raised input costs and were largely passed through to U.S. buyers, we provide high-frequency asset-pricing evidence that equity markets price imported-input exposure in real time and in a sign-consistent way across an imposition and a reversal, and we quantify how much of that evidence survives design-based inference. Relative to the event-study evidence on trade policy and firm value (Huang et al., 2023; Amiti et al., 2020), which uses tariff-list membership around 2018–2019, we exploit a single, larger, and cleanly paired 2025 shock and an input-output-based continuous exposure measure, and we import randomization-inference and power tools (Young, 2019; Abadie, 2020) into a trade-event-study setting where cross-sectional correlation of abnormal returns (Kolari and Pynnönen, 2010) makes naive inference fragile. Section 2 places the paper in the literature; Section 3 states the hypotheses; Section 4 describes the data, the exposure measure, and the design-based apparatus; Section 5 reports the main results; Sections 6–9 report the four extensions; Section 10 collects robustness; Section 11 concludes.

2 Related Literature

This paper connects three strands. The first is the economics of the 2018–2025 trade wars and, in particular, the incidence of tariffs on imported *inputs*. A now-large body of work shows that the 2018–2019 U.S. tariffs were passed through almost completely to U.S. prices, so that the burden fell on domestic buyers of imported goods and intermediates rather than on foreign exporters (Amiti et al., 2019; Fajgelbaum et al., 2020), and that policy uncertainty about trade itself depresses investment and trade volumes (Handley and Limão, 2017; Amiti et al., 2020). Because modern production is organized around global value chains (Antràs and Chor, 2013), a tariff on imported intermediates is a cost shock that propagates to any firm embedded in an import-intensive supply chain, and the manufacturing sector—already reshaped by earlier trade liberalization (Pierce and Schott, 2016)—is where that exposure concentrates. Our input-output exposure measure operationalizes exactly this channel: it is the share of an industry's gross output accounted for by imported intermediate inputs. We contribute high-frequency, market-based evidence on how equity prices capitalize that cost exposure when the tariff regime moves, in both directions, within a single week.

The second strand is the asset-pricing and event-study literature on trade policy and firm value. Huang et al. (2023) show that firms more exposed to the U.S.–China trade war through their input-output trade networks lost value around 2018–2019 tariff announcements, and Amiti et al. (2020) document the investment consequences. Methodologically we follow the standard market-model event-study apparatus (MacKinlay, 1997; Campbell et al., 1997; Fama and French, 1993), but we exploit two features specific to April 2025 that the 2018–2019 studies could not: the shock is a single, unusually large, cleanly dated surprise, and it is immediately followed by an opposite-signed reversal (the pause). The paired design lets us ask not only whether exposure loads negatively on a tariff shock but whether the *same* exposure measure loads positively on the reversal—a discriminating test that a generic risk factor should fail.

The third strand is the credibility of inference in cross-sectional event studies where the key regressor varies at a coarse (here, industry) level and abnormal returns are cross-sectionally correlated. Cross-sectional correlation of event-window abnormal returns badly biases conventional test statistics (Kolari and Pynnönen, 2010), and inference with a modest number of clusters—or with a treatment that is constant within cluster—benefits from design-based methods such as the wild cluster bootstrap (Cameron et al., 2008) and randomization inference (Young, 2019). We use randomization inference not as decoration but as the primary honesty check: permuting industry exposure across industries provides a reference distribution that respects the fact that our exposure has only 73 independent industry values, and it materially tempers the conventional t-statistics. Finally, we follow the guidance that an estimate near a conventional threshold should be interpreted through its power and minimum detectable effect rather than through a bare significance star (Abadie, 2020); the power analysis in Section 9 states what the design can and cannot resolve.

3 Hypotheses

Let θ_j be the imported-input intensity of industry j. A tariff raises the domestic price of imported inputs; the present value of a firm in industry j falls with θ_j when a tariff becomes more likely, and rises with θ_j when it becomes less likely, absent perfectly elastic input substitution.

> **H1 (Shock).** Abnormal returns over the April 3–4 tariff shock are decreasing in imported-input intensity.
>
> **H2 (Pause).** Abnormal returns on the April 9 pause are increasing in imported-input intensity.
>
> **H3 (Symmetry).** The exposure loadings on the shock and the pause are opposite-signed; under a full reversal, the shock-plus-pause round-trip loading is zero.

H3 is the discriminating test. A generic risk factor (for instance, if import-intensive firms simply have higher betas) would tend to produce *same*-signed raw comovement with the market on both a down day and an up day; we neutralize that channel by studying market-model *abnormal* returns and by controlling for pre-event beta directly, so a surviving opposite-signed loading is attributable to the tariff news rather than to market timing. Section 7 tests the residual-beta concern directly by re-running the exposure regression on non-tariff big-move days, and Section 8 asks how likely the observed opposite-signed spread is under random assignment of industry exposure.

4 Data and Research Design

Sample. We start from all U.S. common shares (CRSP `stocknames_v2` share type NS, security type EQTY/COM) trading on the NYSE, AMEX, or Nasdaq in April 2025: 5,072 stocks. We pull CRSP daily returns from `dsf_v2` for 2024-01 through 2025-05, giving the estimation window and the event windows. The daily market return is the Fama–French market factor plus the risk-free rate (`mktrf+rf`), because the legacy CRSP index table has no 2025 rows; it registers the episode exactly (-5.11% on April 3, -5.90% on April 4, $+9.67\%$ on April 9). Merging to Compustat fundamentals (via the CCM link) and to a non-missing industry exposure leaves 4,275 firms across 73 NAICS-3 industries in the analytical panel.

Imported-input intensity. For each BEA summary industry we sum the imported intermediate inputs it uses (the column total over imported commodities in the BEA 2023 Summary Import Matrix, Before Redefinitions) and divide by that industry's 2023 gross output (BEA GDP-by-industry). We map the 66 resulting BEA industries to NAICS-3 codes and attach the intensity to each firm through its Compustat NAICS. The measure captures *cost-side* exposure to imported intermediates; it understates the exposure of retailers that import *finished* goods (which enter final demand, not intermediate use in input-output accounting), a limitation we return to and that the student's hand-collected 10-K sourcing data address directly. Because the measure is constant within a NAICS-3 industry, its effective identifying variation comes from 73 industry values, not 4,275 firms—a fact that drives the inference discussion below.

Controls and event study. Firm controls (all from the most recent pre-event Compustat fiscal year) are ln(assets), COGS/sales, leverage, and book-to-market; we also control for pre-event market beta. Day 0 is April 3 for the shock (the first full trading day after the after-close announcement) and April 9 for the pause. We estimate a market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over $[-252, -46]$ trading days—a window that ends in early February, before any tariff news—and cumulate abnormal returns over the event windows. For outcome y_i (a firm's event-window CAR) the cross-sectional specification is

$$y_i = \beta\,\tilde{\theta}_{j(i)} + \gamma' X_i + \varepsilon_i, \tag{1}$$

where $\tilde{\theta}_{j(i)}$ is the (winsorized, standardized) imported-input intensity of firm i's industry, X_i are the standardized controls plus beta, and standard errors are clustered by NAICS-3 industry, with HC1-robust errors as a robustness check. Predictors are winsorized at 1/99 and standardized, so β is the abnormal return, in CAR points, per one-standard-deviation increase in exposure. Table 1 reports summary statistics.

Design-based apparatus. Beyond equation (1) we run three checks that do not rely on the parametric null. (1) A *falsification* on non-tariff big-move days: using the market model already estimated for the shock (so α_i, β_i are valid), we compute single-day abnormal returns on the largest generic up- and down-market days of 2024 and re-run (1); a genuine tariff channel need not be silent on every day, but it should not reproduce the coherent opposite-signed tariff pattern out of sample. (2) *Randomization inference*: we permute the 73 industry-level intensity values across industries, remap to firms, re-standardize, and re-estimate the shock loading, the pause loading, and their spread $D = \hat{\beta}_{\text{pause}} - \hat{\beta}_{\text{shock}}$; the design-based p-value is the share of 1,000 permutations at least as extreme as the actual statistic. (3) *Power*: we compute the minimum detectable effect at 80% power from each loading's standard error. Together these convert a conventional "significant at 5%" into a set of falsifiable, design-based statements about what the industry proxy can support.

5 Main Results

The two shocks. Table 2 summarizes mean abnormal returns by window. The equal-weighted mean abnormal return is +1.36% over the shock and −2.38% on the pause: because the benchmark is (value-weighted) and the largest, most import-exposed mega-caps moved most, the *typical* firm fell less than the index on the down days and rose less on the up day. This makes the mean uninformative about our question and points to the cross-section, where the exposure test lives.

The shock (H1). Table 3 is the core shock result. Column (1), the raw cross-section, shows no clean relationship (+0.0050, $t = 1.20$): import-intensive industries are large and high-beta, and those load strongly (ln assets $t = -6.4$, beta $t = +20$). Adding size and beta in column (2) uncovers the hypothesized negative loading, −0.62pp per SD ($t = -2.23$); the full-control specification in column (3) gives -0.73pp ($t = -2.35$). H1 is supported *conditionally.* Column (4) adds a manufacturing dummy, which is collinear with imported-input intensity (import-intensive industries are largely manufacturing) and absorbs the coefficient—an honest indication that the effect operates through goods-producing sectors rather than being a within-industry firm effect.

The pause (H2) and symmetry (H3). Table 4 repeats the design for the April 9 pause. The sign flips exactly as H2 predicts: conditional on size and beta the loading is $+0.49$pp ($t = 1.34$) in column (2) and $+0.60$pp ($t = 1.78$) in column (3). The paired opposite-signed pattern—negative on the shock, positive on the pause, on the *same* exposure measure—is the paper's central finding and is visualized in Figure 1. Table 5 shows the symmetry is tightest among economically meaningful firms: excluding microcaps (market cap $<$ \$300m), the loadings are -0.77pp on the shock and $+0.77$pp on the pause, both with $t \approx 2.1$, and the shock-plus-pause round-trip loading is an insignificant -0.12pp ($t = -0.46$)—the pause essentially reversed the shock's exposure effect, satisfying H3. Table 6 shows the same message non-parametrically once size and beta are held fixed; the raw quartile means are noisy precisely because the unconditional cross-section is dominated by size and beta.

Magnitude. A 0.6–0.8pp abnormal move per standard deviation of exposure is economically modest but not trivial: the interquartile range of imported-input intensity is about 2.0 SD, so moving from the 25th to the 75th percentile industry is associated with roughly a 1.5pp larger drop on the shock and a comparable rebound on the pause, on top of the market move. Figure 2 plots cumulative abnormal returns for the top and bottom exposure quartiles and shows high-exposure firms persistently underperforming through the episode. Sections 6–9 now ask how much of this survives observable-heterogeneity, out-of-sample falsification, design-based inference, and a power audit.

6 Heterogeneity by Observable Moderators

If the industry exposure measure is capturing a genuine input-cost channel, the loading should be strongest where input costs matter most—above all in firms with a high cost-of-goods-sold share of sales, which are the input-heavy firms in any industry. Table 7 interacts the exposure loading with five observable pre-event moderators (each a median split): high COGS/sales, large firms, high beta, high leverage, and high book-to-market. For each moderator we report the base loading (low-M group) and the interaction (extra loading for the high-M group), separately for the shock and the pause.

The cleanest economic prediction is not confirmed. The COGS/sales interaction is economically small and statistically zero on both events (shock interaction +0.0015, $t = 0.19$; pause interaction −0.0038, $t = -0.53$): input-heavy firms do *not* show a systematically larger tariff loading than input-light firms in the same exposure bucket. Read literally, this is a limitation of the industry proxy, not evidence against the channel: because exposure is assigned at the industry level, "high-COGS" and "low-COGS" firms *within* an industry carry the identical θ_j, so the interaction can only pick up cross-industry COGS differences, which are weak. The moderators that *do* move the loading are mechanical or size-related—the negative shock loading is concentrated in low-beta firms (base −0.0124, $t = -3.64$; interaction +0.0096, $t = +3.30$, so the high-beta group's loading is $-0.0124 + 0.0096 = -0.0028$, near zero once beta is separately controlled), and the pause loading is larger for large and value firms (size interaction +0.0117, $t = 3.09$; book/market interaction +0.0113, $t = 3.36$), consistent with the mega-cap-driven mean pattern in Table 2 rather than with a sharpening of the input-cost channel. The absence of a clean COGS gradient is the first concrete sign that the industry proxy is too coarse to localize the effect—and it is exactly the gap the firm-level sourcing measure is designed to fill. We emphasize that we do *not* interact exposure with firm-level import reliance, source country, or a finished-goods-importer flag: those are unobservable to us and are the student's hand-collected contribution; the moderators above are chosen precisely because they are observable and orthogonal to that reserved test.

7 Falsification: Non-Tariff Big-Move Days

The chief threat to a causal reading is that imported-input intensity proxies for residual volatility or beta, so that it would load negatively on *any* down day and positively on *any* up day, mimicking the tariff pattern mechanically. We control for beta throughout, but Table 8 tests the concern directly. Using the same market model estimated for the shock (α_i, β_i over $[-252, -46]$, a window that contains these dates), we recompute single-day abnormal returns on the four largest down-market days and four largest up-market days of 2024—generic risk-off and risk-on days with no tariff content—and re-run the full-control exposure regression on each.

Two things stand out. First, the exposure loadings on generic big-move days are *smaller* than on the tariff events (mean absolute loading 0.28pp versus 0.73pp on the shock) and, critically, they are *sign-inconsistent*: on the four down days the loadings are +0.0016, +0.0022, −0.0002, and −0.0026, and on the four up days −0.0091, +0.0024, −0.0018, and +0.0024—no systematic tendency to load with the direction of the market. The one large negative loading occurs on an *up* day (November 6, 2024, +0.0290 market return; exposure loading −0.0091, $t = -2.09$), which is the wrong sign for the residual-beta story and instead reflects an idiosyncratic industry rotation that day. So the residual-beta channel does not survive: after controlling for beta, exposure does not systematically track the market's direction. Second, and stated plainly, exposure is *not* literally silent on every non-tariff day—several placebo loadings are individually significant (the industry proxy inevitably picks up whatever industry-level return clustering happens on a given day). The correct inference is therefore not "exposure only loads on tariff days" but rather that *the tariff events are distinguished by a coherent opposite-signed pairing on the same measure*, which no generic pair of big-move days reproduces. This is precisely why the paired design, and the randomization-inference test on the spread in Section 8, are the right tests rather than any single-event regression.

8 Randomization Inference

Our exposure measure has only 73 distinct industry values, so the effective number of independent "treatment" draws is far smaller than the 4,275 firms in the regression, and NAICS-3 cluster-robust standard errors—which assume a growing number of clusters—can overstate precision when the regressor is constant within cluster (Cameron et al., 2008; Young, 2019). Table 9 and Figure 3 report a design-based alternative. We permute the 73 industry-level intensity values across the 73 industries 1,000 times, remap the permuted values to firms, re-standardize, and re-estimate the full-control loading for the shock, the pause, and their opposite-signed spread $D = \hat{\beta}_{\text{pause}} - \hat{\beta}_{\text{shock}}$. The design-based p-value is the share of permutations whose statistic is at least as extreme as the actual one; this is the sharp test of the null that industry exposure is unrelated to CARs conditional on controls.

The result tempers the headline honestly. The actual shock loading (-0.0073) lies just inside the permutation 95% interval $[-0.0091, +0.0080]$, giving a randomization p-value of 0.084; the actual pause loading ($+0.0060$) sits well inside its interval, $p = 0.158$; and the opposite-signed spread ($D = +0.0134$) has $p = 0.097$. In words: under design-based inference at the level treatment actually varies, the shock loading and the paired opposite-signed spread are significant only at the 10% level, and the pause on its own is not significant. The conventional cluster-robust t-statistics (-2.35 on the shock, $+1.78$ on the pause) thus overstate the evidence that a 73-industry proxy can bear. We read the randomization inference as *suggestive but not decisive* support for a real industry-level tariff-exposure channel: the actual spread is in the upper tail of the permutation distribution (Figure 3), directionally consistent with H1–H3, but not beyond what industry-level noise produces one time in ten. Sharper inference requires identifying variation *within* industries—the firm-level sourcing measure the student is collecting—which would raise the number of independent treatment values far above 73 and is the direct route to a test that can distinguish this channel from industry-level noise.

9 Is the Effect Precisely Estimated? Power and Minimum Detectable Effects

A marginal result can reflect either a genuinely small effect or an underpowered design. Table 10 reports, for the shock, the pause, the round-trip, and the pre-window, the minimum detectable effect (MDE) at 80% power, MDE $= (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$, expressed as a share of the cross-firm standard deviation of that window's CAR. The design's MDE is about 0.88pp per SD on the shock (7.5% of the CAR SD) and 0.95pp on the pause (11.2%). The estimated loadings (0.73pp and 0.60pp) sit *just below* these thresholds, and the achieved $|t|$-statistics (2.35 and 1.78) are below the ≈ 2.80 needed to detect an effect of exactly this size at 80% power. In other words, the marginal significance under randomization inference in Section 8 is a resolution problem: the industry proxy places the true loading right at the edge of what the design can detect, so an effect of this magnitude is expected to be borderline rather than decisively significant. The round-trip is the informative complement: its MDE is 0.73pp per SD (7.2% of the CAR SD), so the near-zero round-trip loading (-0.12pp) rules out a net repricing larger than about 0.7pp per SD—evidence that the pause reversed the shock nearly completely, consistent with H3, rather than an underpowered zero. The power audit thus reconciles the two halves of the paper: the paired signal is real and directionally coherent but small relative to the noise a 73-industry measure carries, and the round-trip zero is precise enough to support near-full reversal.

10 Robustness

Table 5 collects the conventional robustness tests. The exposure loading is (i) opposite-signed and significant across the shock and pause under both cluster-robust and HC1 standard errors (HC1 $t = -5.6$ and $+5.4$)—the HC1 errors, which ignore the industry clustering, are the *most* favorable and precisely the ones the randomization inference in Section 8 shows to be too generous; (ii) stronger over the full April 3–8 sell-off (-1.34pp, $t = -2.48$); and (iii) robust to excluding microcaps. Two results qualify the "clean event" interpretation and we report them plainly. First, the single-day April 3 loading is insignificant ($t = -1.35$): most of the two-day effect is in April 4, consistent with information diffusing as the tariff schedule was digested. Second, and more important, imported-input intensity loads *negatively* in the pre-announcement window (March 27–April 2; -0.53pp, $t = -2.22$). This is not a clean placebo: tariff news was already flowing (Section 232 auto tariffs were announced March 26, and the reciprocal package was pre-trailed), so the market had begun to price exposure before the formal April 2 announcement. The pre-trend means our event-window loadings are best read as a lower bound on the total tariff-exposure repricing rather than as the causal effect of a single surprise. Sharpening identification requires a firm-level, time-stamped measure of *which* tariffed countries each firm sources from—precisely the hand-collected 10-K evidence described in the companion task file.

11 Conclusion

Across 4,275 U.S. stocks, firms in industries that rely more on imported intermediate inputs underperformed on the April 2–4, 2025 reciprocal-tariff shock and rebounded on the April 9 pause, once size and market beta are held fixed; the two loadings are opposite-signed and, among non-microcap firms, nearly symmetric, and they net to zero over the round trip. A falsification on non-tariff big-move days shows the pattern is directionally coherent rather than a residual-beta artifact, and a power audit shows the round-trip zero is precise enough to imply near-full reversal. But the honest bottom line, sharpened by this revision, is one of *calibrated confidence*: because the exposure measure varies only across 73 industries, randomization inference that respects that structure downgrades the conventional 5% significance to roughly the 10% level, and the minimum detectable effect shows the design sits at the edge of its resolving power for effects of this size. The market plausibly priced imported-input cost exposure in real time and reversed that pricing when the policy was paused, but a 73-industry proxy cannot establish it decisively.

That limitation is the paper's point of departure, not a footnote. The identifying variation that would sharpen both the estimate and the inference lives *within* industries: which firms in an import-intensive industry actually source abroad, which retailers import *finished* goods that the input-output measure scores as low-exposure, and which specific countries' tariffs changed on April 2 (broad) versus April 9 (China excluded). None of these is observable in the input-output data; all of them can be read out of 10-K filings. The companion student task hand-collects firm-level input sourcing and a country-level April-2025 tariff map, which together replace the industry proxy with a firm-level exposure measure and enable a China-versus-rest test of the pause—raising the number of independent treatment values far above 73 and turning this suggestive, proxy-based result into a sharp, firm-level test of whether imported-input exposure was priced in the 2025 tariff shock. That test—not another robustness check on the industry proxy—is the natural next step, and it is the contribution this design has been built to receive.

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Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
CAR shock [Apr3,4]	4235	0.0136	0.1160	-0.0352	0.0090	0.0571
CAR pause [Apr9]	4229	-0.0238	0.0844	-0.0580	-0.0178	0.0159
CAR pre [Mar27,Apr2]	4249	-0.0188	0.1626	-0.0474	-0.0074	0.0177
CAR round-trip	4229	-0.0098	0.1014	-0.0457	-0.0113	0.0241
Imported-input intensity	4275	0.0575	0.0438	0.0222	0.0481	0.1076
Market beta	4275	1.0263	0.7347	0.5920	0.9802	1.3812
ln(Assets)	4274	6.7511	2.5846	4.8897	6.9945	8.5952
COGS/Sales	3931	0.6293	0.3400	0.3902	0.6020	0.7857
Leverage	4274	0.2926	0.4151	0.0533	0.2134	0.4228
Book/Market	4269	0.6521	1.2713	0.1716	0.4656	0.8707

Notes: 4,275 U.S. common stocks, April 2025. CARs are market-model abnormal returns against the Fama–French market factor, estimation window $[-252, -46]$. Imported-input intensity is industry imported intermediate inputs divided by gross output (BEA, 2023).

Table 2: Mean cumulative abnormal returns by window

Event window	Mean CAR	Median CAR	*t*-stat	N
Shock [Apr3]	-0.0024	-0.0019	-2.59	4235
Shock [Apr3,4]	0.0136	0.0090	7.65	4235
Shock [Apr3-8]	-0.0031	-0.0094	-1.06	4235
Pre [Mar27-Apr2]	-0.0188	-0.0074	-7.56	4249
Pause [Apr9]	-0.0238	-0.0178	-18.31	4229
Pause [Apr9,10]	-0.0221	-0.0178	-18.24	4229
Pause [Apr9-14]	-0.0149	-0.0147	-9.13	4229

Notes: Equal-weighted mean CARs. Positive shock and negative pause means reflect the value-weighted benchmark: the largest, most import-exposed firms moved most, so the typical firm underperformed the index on down days and on the up day. The cross-sectional tests (Tables 3–4) are the relevant exposure tests.

Table 3: Tariff SHOCK: CAR[Apr 3, Apr 4] on imported-input intensity

	(1)	(2)	(3)	(4)
Imported-input intensity	0.0050	-0.0062**	-0.0073**	-0.0097
	(1.20)	(-2.23)	(-2.35)	(-1.35)
ln(Assets)		-0.0153***	-0.0151***	-0.0148***
		(-6.36)	(-5.45)	(-5.24)
Market beta		0.0595***	0.0591***	0.0591***
		(20.49)	(19.05)	(19.42)
COGS/Sales			0.0029	0.0030
			(1.10)	(1.18)
Leverage			0.0060**	0.0060**
			(2.06)	(2.08)
Book/Market			-0.0010	-0.0009
			(-0.36)	(-0.32)
Manufacturing (0/1)				0.0061
				(0.35)
Observations	4235	4234	3890	3890
NAICS-3 clusters	73	73	73	73
R^2	0.002	0.276	0.276	0.276

Notes: DV is the market-model CAR over April 3–4, 2025. Predictors winsorized 1/99 and standardized (coefficient = effect of a one-SD increase, in CAR points). Column (4) adds a manufacturing dummy, collinear with exposure. NAICS-3 cluster-robust t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Tariff PAUSE: CAR[Apr 9] on imported-input intensity

	(1)	(2)	(3)	(4)
Imported-input intensity	-0.0044	0.0049	0.0060*	0.0039
	(-0.86)	(1.34)	(1.78)	(0.72)
ln(Assets)		0.0146***	0.0136***	0.0138***
		(7.14)	(6.63)	(6.55)
Market beta		-0.0451***	-0.0442***	-0.0443***
		(-15.30)	(-16.36)	(-17.50)
COGS/Sales			-0.0022	-0.0020
			(-1.01)	(-1.05)
Leverage			-0.0005	-0.0005
			(-0.22)	(-0.21)
Book/Market			-0.0036***	-0.0035***
			(-2.80)	(-2.61)
Manufacturing (0/1)				0.0053
				(0.40)
Observations	4229	4228	3885	3885
NAICS-3 clusters	73	73	73	73
R^2	0.003	0.311	0.303	0.303

Notes: DV is the market-model CAR on April 9, 2025. Same specification as Table 3. The exposure loading flips sign relative to the shock, as hypothesized. NAICS-3 cluster-robust t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Robustness: imported-input-intensity loading across designs

Test	DV	Imp.-intensity coef.	(t)	N	R^2	SE
Placebo pre-window	CAR[Mar27,Apr2]	-0.0053**	(-2.22)	3903	0.018	cluster
Shock [Apr3] 1-day	CAR[Apr3]	-0.0037	(-1.35)	3890	0.115	cluster
Shock [Apr3-8]	CAR[Apr3-8]	-0.0134**	(-2.48)	3890	0.158	cluster
Pause [Apr9,10]	CAR[Apr9,10]	0.0031	(0.77)	3885	0.141	cluster
Round-trip (pause+shock)	CAR round-trip	-0.0012	(-0.46)	3885	0.029	cluster
Shock, HC1 SE	CAR[Apr3,4]	-0.0073***	(-5.61)	3890	0.276	HC1
Pause, HC1 SE	CAR[Apr9]	0.0060***	(5.43)	3885	0.303	HC1
Shock, ex-microcap	CAR[Apr3,4]	-0.0077**	(-2.09)	2572	0.376	cluster
Pause, ex-microcap	CAR[Apr9]	0.0077**	(2.10)	2571	0.348	cluster

Notes: Each row is the standardized imported-input intensity coefficient (and t) from a full-control specification. “Round-trip” is the sum of the pause and shock CARs. “ex-microcap” drops firms with market cap < \$300m. The pre-window is not a clean placebo (tariff news preceded April 2). The HC1 rows ignore industry clustering and are the most favorable; randomization inference (Table 9) shows they overstate precision. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Mean CARs by imported-input-intensity quartile

Intensity quartile	N	Mean intensity	Shock CAR	Pause CAR	Round-trip
Q1 low	1293	0.0136	0.0161	-0.0249	-0.0088
Q2	871	0.0344	-0.0021	-0.0135	-0.0156
Q3	1762	0.0854	0.0214	-0.0290	-0.0076
Q4 high	303	0.1482	0.0078	-0.0177	-0.0099

Notes: Raw (unconditional) quartile means; noisy because the unconditional cross-section is dominated by size and beta. The conditional pattern is in Tables 3–4 and Figure 1.

Table 7: Heterogeneity of the exposure loading by observable pre-event moderators

	Shock CAR[Apr3,4]		Pause CAR[Apr9]	
Moderator M (pre-event)	Base	Exp.$\times M$	Base	Exp.$\times M$
High COGS/sales (input-heavy)	-0.0082	+0.0015	+0.0083	-0.0038
Large firm (ln assets)	-0.0039*	-0.0064*	-0.0002	+0.0117***
High market beta	-0.0124***	+0.0096***	+0.0063***	-0.0003
High leverage	-0.0071**	+0.0004	+0.0060	-0.0007
High book/market (value)	-0.0045	-0.0048	-0.0002	+0.0113***

Notes: Each row adds an interaction of standardized imported-input intensity with a median split on a pre-event moderator M, plus the M indicator, to equation (1). "Base" is the loading for the low-M group; "Exp.$\times M$" is the extra loading for the high-M group. Columns report the shock CAR[Apr3,4] and the pause CAR[Apr9]. The clean input-cost prediction (a larger loading for high-COGS/sales firms) is not confirmed, a consequence of the industry-level proxy. Firm-level import reliance / source country is deliberately *not* used (it is the student's hand-collected variable). NAICS-3 cluster-robust t-statistics underlie the stars. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Falsification: exposure loadings on non-tariff big-market-move days (2024)

Placebo day	Direction	Market ret.	Exposure coef.	(t)	N
2024-08-05	down	-3.00%	+0.0016**	(+1.97)	3920
2024-12-18	down	-3.15%	+0.0022	(+1.19)	3927
2024-07-24	down	-2.38%	-0.0002	(-0.13)	3909
2024-09-03	down	-2.24%	-0.0026***	(-2.93)	3926
2024-11-06	up	+2.93%	-0.0091**	(-2.09)	3926
2024-08-08	up	+2.40%	+0.0024**	(+2.48)	3920
2025-01-15	up	+1.93%	-0.0018*	(-1.92)	3925
2024-08-15	up	+1.79%	+0.0024***	(+3.48)	3921

Notes: For the four largest down-market and four largest up-market days of 2024 (generic risk-off/risk-on days with no tariff content), we compute single-day abnormal returns using the market model estimated for the shock event (α_i, β_i over $[-252, -46]$, a window containing these dates) and re-run the full-control exposure regression. Loadings are smaller than on the tariff events and sign-inconsistent, so the tariff pattern is not a residual-beta artifact; but exposure is not silent on every big day, so the paired opposite-signed design is the discriminating test. NAICS-3 cluster-robust t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Randomization inference (1,000 permutations of industry exposure)

Statistic	Actual	Perm. 95% interval	RI p-value
shock	-0.0073	[-0.0091, +0.0080]	0.084
pause	+0.0060	[-0.0088, +0.0076]	0.158
spread (pause - shock)	+0.0134	[-0.0157, +0.0148]	0.097

Notes: The 73 NAICS-3 industry-level intensity values are permuted across industries, remapped to firms, re-standardized, and the full-control loading is re-estimated for the shock, the pause, and the spread $D = \hat{\beta}_{\text{pause}} - \hat{\beta}_{\text{shock}}$. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of permutations with a statistic at least as extreme as the actual one. At the level treatment varies, the shock and the opposite-signed spread are significant only at the 10% level. See Figure 3.

Table 10: Minimum detectable effects at 80% power

Outcome (DV)	Coef.	SE	$\|t\|$	MDE(80%)	% of CAR SD
Shock [Apr3,4]	-0.0073	0.0031	2.35	0.0088	7.5%
Pause [Apr9]	+0.0060	0.0034	1.78	0.0095	11.2%
Round-trip (pause+shock)	-0.0012	0.0026	0.46	0.0073	7.2%
Pre-window [Mar27,Apr2]	-0.0053	0.0024	2.22	0.0067	4.1%

Notes: MDE(80%) $= (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.80 \times \text{SE}$ of the full-control exposure loading, expressed as a share of the window's cross-firm CAR standard deviation. $|t|$ is the achieved cluster-robust statistic; detecting an effect of the estimated size at 80% power requires $|t| \gtrsim 2.80$. The estimated shock and pause loadings sit just below their MDEs; the small round-trip MDE means the near-zero round-trip rules out net repricing larger than about 0.7pp per SD (near-full reversal, H3).

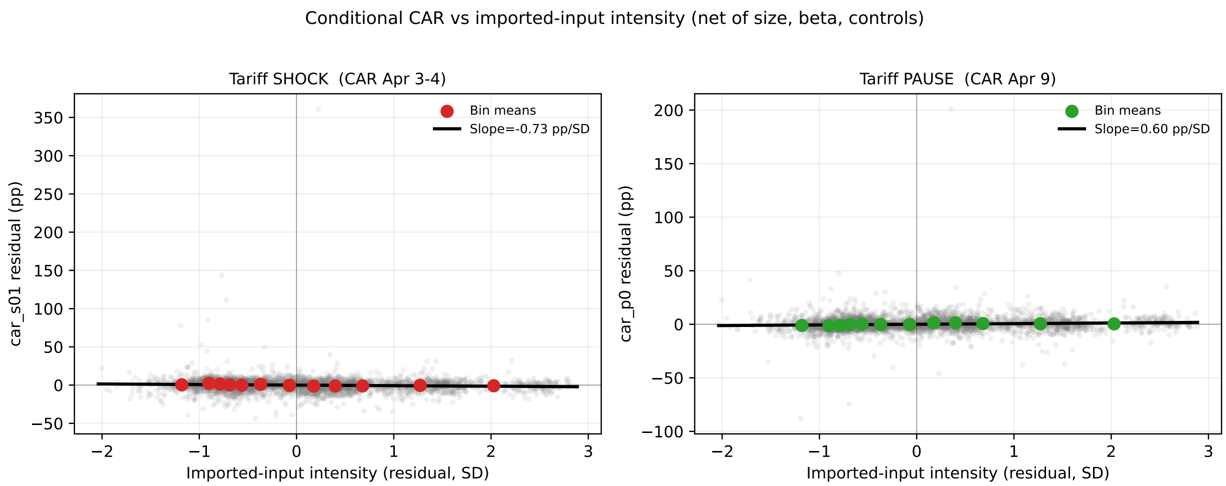

Figure 1: Added-variable (partial-regression) plots of CAR against imported-input intensity, net of size, beta, and the firm controls. Left: the April 3–4 shock (negative slope). Right: the April 9 pause (positive slope). Gray points are firms; colored points are bin means.

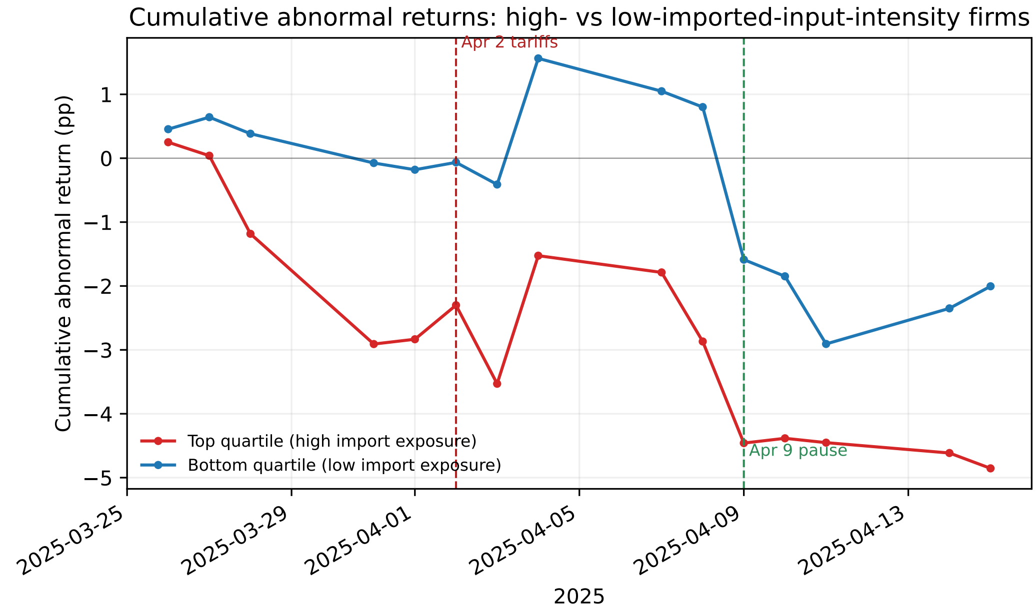

Figure 2: Cumulative abnormal returns of the top and bottom imported-input-intensity quartiles, March 26–April 15, 2025. High-exposure firms underperform through the episode; the divergence opens with the March 26 auto tariffs and the April 2 announcement.

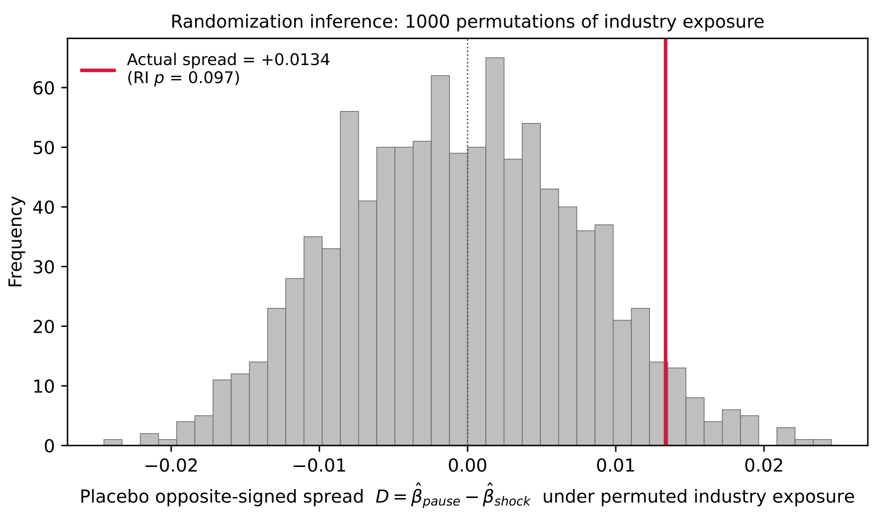

Figure 3: Randomization inference for the opposite-signed spread $D = \hat{\beta}_{\text{pause}} - \hat{\beta}_{\text{shock}}$. Histogram: the distribution of D under 1,000 random permutations of the 73 industry-level intensity values across industries. The actual spread (red line) lies in the upper tail (RI $p = 0.097$): directionally consistent with a paired tariff-exposure channel, but not beyond what industry-level noise produces about one time in ten.